RMT: Kernel Spectral Clustering with Sparsity

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1 Introduction

Consider n observations $x_1, \cdots, x_n$ in dimension d which we want to split into groups. For example, the x_i's could represent images of animals which we want to group by species. Here, we consider a simplified model where the x_i's are assumed to form two clusters symmetric about the origin. Specifically, we define a vector $y \in \{-1, +1\}^n$ such that $x_i \sim \mathcal{N}(+\mu, Id)$ if $y_i = +1$ and $x_i \sim \mathcal{N}(-\mu, Id)$ if $y_i = -1$.

Question 1 Write the $d \times n$ matrix $X = [x_1 \cdots x_n]$ as a sum of a rank-one deterministic matrix and a random matrix Z such that $\mathbb{E}[Z] = 0_{d\times n}$. How is this type of model commonly called?

Answer 1 We can write $x_i = y_i\mu + n_i$ with $n_i \sim \mathcal{N}(0, Id)$. Thus with $Z = [n_1 \cdots n_n]$ and $A = [y_1\mu \cdots y_n\mu] = \mu y^T$, $X = Z + A$, $\mathbb{E}[Z] = 0$ and $rk(A) = 1$ if $\mu \neq 0$. Indeed every vector of A are collinear. This type of model is called a **spiked model**.

Our goal is to reconstruct y from the Gram kernel matrix $K = \frac{1}{d}X^TX$. We consider this problem in the large-dimensional regime modeled by $d, n \to +\infty$ with $0 < c := d/n < +\infty$. In the following, we will simply write "$d, n \to +\infty$".

Question 2 Recall how this large-dimensional regime relates to practical situations. What limitations do you expect regarding the size of K?

Answer 2 In practice, we do have a large number of observations and dimensions (features), which justifies placing ourselves in an asymptotic regime, with a certain ratio $c = d/n$. Regarding the size of $K \in \mathcal{M}_n(\mathbb{R})$, this could a be a very huge but almost sparse matrix. It is therefore necessary for example to use a threshold to 0 in order to store it efficiently.

To circumvent such limitations we propose to compute only a portion $\epsilon \in]0, 1]$, chosen at random, of the entries of K. Specifically, we consider instead $K_\epsilon = K \odot B$ where $\odot$ denotes the Hadamard (element-wise) product and B is an $n \times n$ symmetric random matrix with i.i.d. Bernoulli(ϵ) non-diagonal entries and all diagonal entries equal to 1.

2 Limiting spectral behavior

In the following, we are going to study a spectral clustering algorithm based on K_ϵ in order to quantify the impact of the sparsity level ϵ on the clustering performance. Define the resolvent of K_ϵ as $Q(z) = (K_\epsilon - zI_n)^{-1}$ for all $z \in \mathbb{C}$ which is not an eigenvalue of K_ϵ. We accept the following result.

Claim 1 For $z \in \mathbb{C}^+ = \{z \in \mathbb{C}, \quad \operatorname{Im} z > 0\}$, define the $n \times n$ deterministic matrix :

$$\bar{\mathbf{Q}}(z) = \bar{m}(z)\left(I_n + \|\mu\|^2 \frac{\epsilon\bar{m}(z)}{c + \epsilon\bar{m}(z)} \frac{yy^T}{n}\right)^{-1} \tag{1}$$

Where $\bar{m}$ is the unique Stieltjes transform solution to:

$$\left(1 - z - \frac{\epsilon}{c}\bar{m}(z) + \frac{\epsilon^3}{c}\frac{\bar{m}(z)^2}{c + \epsilon\bar{m}(z)}\right)\bar{m}(z) = 1 \tag{2}$$

$\bar{\mathbf{Q}}$ is a deterministic equivalent of the (random) resolvent $\mathbf{Q}$ in the sense that, for all (sequences of) vectors $a, b \in \mathbb{R}_n$ with unit Euclidean norm and matrices $A \in \mathbb{R}_{n\times n}$ with unit spectral norm, the following almost-sure convergences are true for all $z \in \mathbb{C}^+$,

$$a^T Q(z)b - a^T \bar{Q}(z)b \underset{d,n\to+\infty}{\xrightarrow{a.s.}} 0 \tag{3}$$

$$\frac{1}{n}Tr(AQ(z)) - \frac{1}{n}Tr(A\bar{Q}(z)) \underset{d,n\to+\infty}{\xrightarrow{a.s.}} 0 \tag{4}$$

Question 3 Find a degree-3 polynomial P_ϵ such that $P_\epsilon(\bar{m}(z)) = 0$. Which equation do you recognize if $\epsilon = 1$? Is it expected?

Answer 3 Using eq. (1), we find :

$$P_\epsilon(m) = (1-\epsilon)\frac{\epsilon^2}{c^2}m^3 + z\frac{\epsilon}{c}m^2 + \left[(z-1) + \frac{\epsilon}{c}\right]m + 1$$

Then $P_\epsilon(\bar{m}(z)) = 0$.

If $\epsilon = 0$ then we have $(1-z)\bar{m}(z) = 1$, which is expected as then $K_0 = Diag(\|x_i\|_2/d) \approx I_n$ (strong law of large numbers and $\mathbb{E}\, n^2 = 1$ if $n\mathcal{N}(0,1)$), $Q = (K_0 - zI_n)^{-1} \approx \frac{1}{1-z}I_n$, thus $\bar{m}(z) = \frac{1}{1-z} = \mathcal{ST}(\delta_1)(z)$.

And if $\epsilon = 1$, we find $\frac{z}{c}m^2 + [z + (\frac{1}{c} - 1)]m + 1 = 0$, which is the polynomial whose solution is the Stieltjes transform of the Marchenko-Pastur distribution, of variance 1 and ratio $c \leftarrow \frac{1}{c}$. Indeed, with $\epsilon = 1$, $K_1 = K$. And our model then fits into the assumptions of Marchenko-Pastur theorem with $p \leftarrow n$ and $n \leftarrow d$ and $\frac{1}{n}XX^\star \leftarrow K = \frac{1}{d}X^TX$ (ignoring the spike effect, which does not affect the distribution but rather the extreme values).

Question 4 Thanks to the change of variables $\tilde{z} = \sqrt{\frac{c}{\epsilon}}(z-1)$ and $\tilde{m}(\tilde{z}) = \sqrt{\frac{\epsilon}{c}}\bar{m}(z)$, describe the behavior of the limiting spectral distribution of K_ϵ as $\epsilon \to 0$.

Answer 4 Using this change of variable we have $\tilde{m}$ solution of $(1-\epsilon)\sqrt{\frac{\epsilon}{c}}\tilde{m}^3 + (\sqrt{\frac{\epsilon}{c}}\tilde{z} + 1)\tilde{m}^2 + [\tilde{z} + \sqrt{\frac{\epsilon}{c}}]\tilde{m} + 1 = 0$.

But $\tilde{m}(\tilde{z}) = \sqrt{\frac{\epsilon}{c}}\bar{m}(z) = \mathcal{O}_z\left(\sqrt{\frac{\epsilon}{c}}\right)$ and $\tilde{z} = \sqrt{\frac{c}{\epsilon}}(z-1)$, thus:

$$-(\tilde{z}\tilde{m} + 1) = (1-\epsilon)\sqrt{\frac{\epsilon}{c}}\tilde{m}^3 + \tilde{m}^2 + \sqrt{\frac{\epsilon}{c}}\tilde{z}\tilde{m}^2 + \sqrt{\frac{\epsilon}{c}}\tilde{m} = \mathcal{O}_z\left(\frac{\epsilon}{c}\right)$$

Finally:

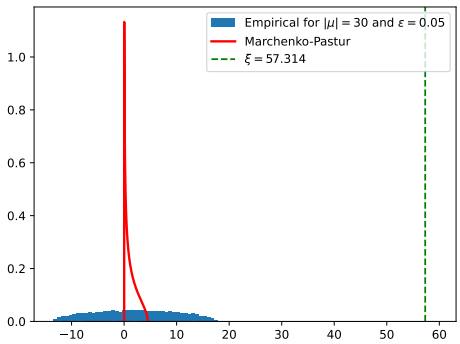
$$\tilde{m} = -\frac{1}{\tilde{z}} + \mathcal{O}_z(\epsilon^{\frac{3}{2}}), \quad \bar{m} = -\sqrt{\frac{c}{\epsilon}}\frac{1}{\tilde{z}} + \mathcal{O}_z(\epsilon) = \frac{1}{1-z} + \mathcal{O}_z(\epsilon)$$

And $\bar{m}_\epsilon \underset{\epsilon\to 0}{\to} \bar{m}_0 = \mathcal{ST}(\delta_1)$. Thus we have weak convergence of the associated measure. And the limiting spectral distribution of K_ϵ is δ_1, which is expected as $K_0 = Diag(\|x_i\|_2/d) \approx I_n$.

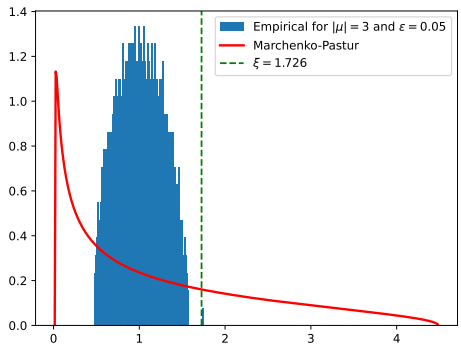
Question 5 For various values of ε and $\|\mu\|$, plot the histogram of eigenvalues of K_ε with the density of its limiting spectral distribution. Comment on the shape of the spectrum. Elaborate on the specific cases $\varepsilon = 1$ and $\varepsilon \to 0$ in light of your previous answers. What is the influence of $\|\mu\|$ on the spectrum of K_ε ? Discuss this phenomenon.

Answer 5 Indeed, for values of ε that get close to 0, the distribution gets closer and closer to the dirac in 1 (Figure 1). However, the distribution is stretched for high values of $||\mu||$, and we can notice that it is approximately the distribution of the half-circle for small ε, and Marchenko-Pastur for ε around 1. High values of $||\mu||$ also send the spike eigenvalue very far from the bulk.

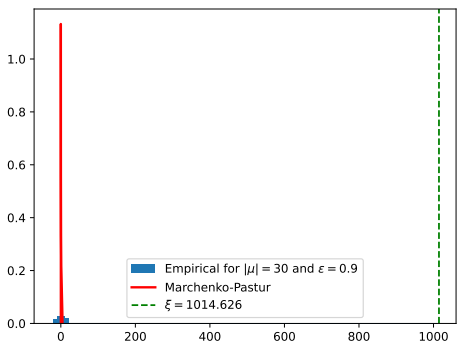
Question 6 Observe and comment on the shape of an eigenvector associated with the largest eigenvalue for various values of ϵ and $\|\mu\|$. Propose an algorithm to cluster the x_i's. In your simulations, it may help visualization to sort the x_i's such that $x_1, \cdots, x_{n_0}$ belong to the first group and $x_{n_0+1}, \cdots, x_n$ belong to the second group.



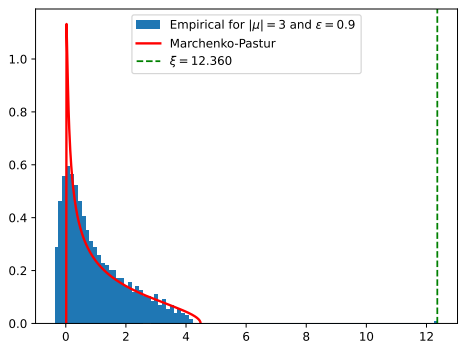
(a) $\varepsilon << 1, ||\mu|| >> 1$



(b) $\varepsilon << 1, ||\mu|| \approx 1$

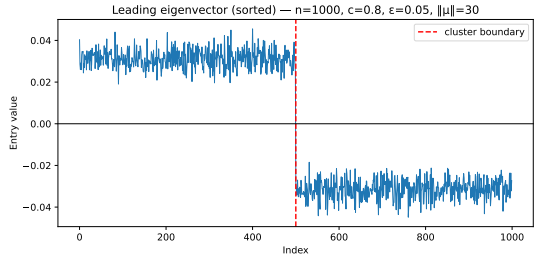


(c) $\varepsilon \simeq 1, ||\mu|| >> 1$

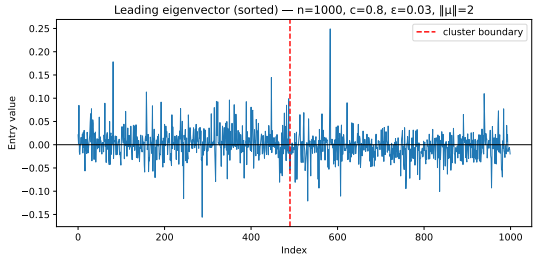


(d) $\varepsilon \simeq 1, ||\mu|| \approx 1$

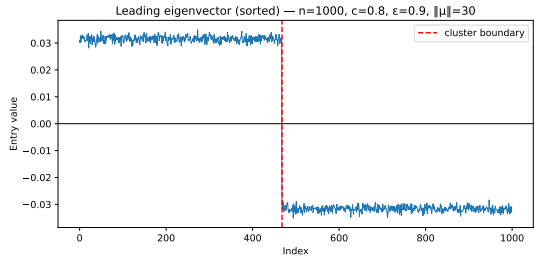
Figure 1: Spectrum of K_ϵ



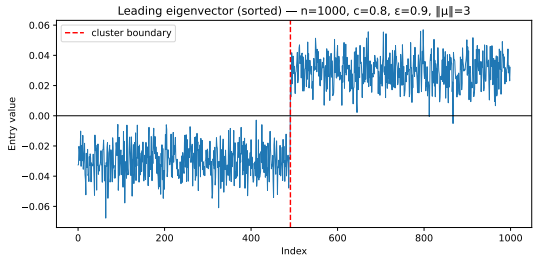
(a) $\varepsilon << 1, ||\mu|| >> 1$



(b) $\varepsilon << 1, ||\mu|| \approx 1$



(c) $\varepsilon \simeq 1, ||\mu|| >> 1$



(d) $\varepsilon \simeq 1, ||\mu|| \approx 1$

Figure 2: Eigenvectors of K_ϵ

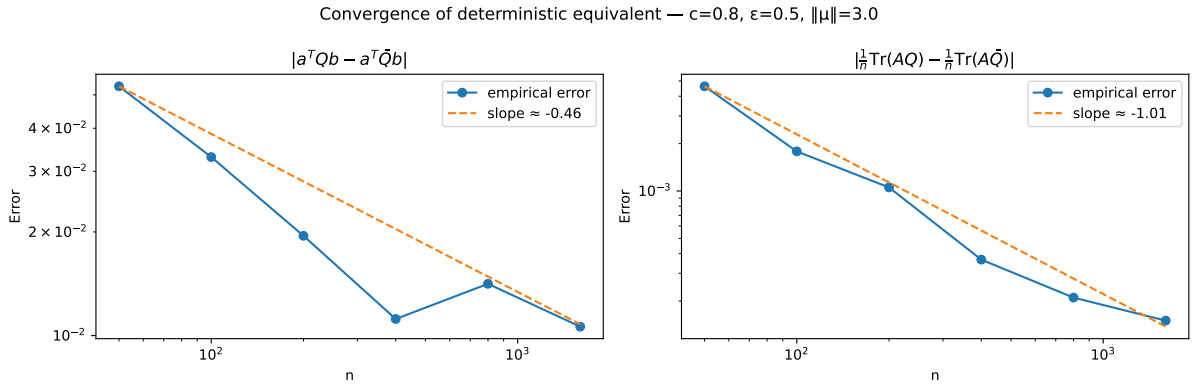


Figure 3: Almost-sure convergence

Answer 6 We simply use an analog of spectral clustering: take the sign of the leading eigenvector's entries as the predicted label (Figure 2). This clearly separate the indices in two part of opposite sign. Note that for ϵ and $||\mu||$ very small, there is a phase change and the signal becomes pure noise: The clustering algorithm fails.

Question 7 For growing values of n, verify with simulations the convergences (3) and (4). Plot your results. Can you guess the speeds of convergence?

Answer 7 We can easily guess the speed of convergence using the slope in the log-log scale: For 3, the convergence should be in $\frac{1}{\sqrt{n}}$ and for 4 it should be in $\frac{1}{n}$ (Figure 3).

3 The dominant eigenvalue and its eigenvector

Denote λ_1 the dominant eigenvalue of K_ϵ and u_1 a unit-norm eigenvector associated to λ_1.

Question 8 Given your previous observations and your knowledge on similar models, what can you expect from the asymptotic behavior of λ_1? Relying on Claim 1, make a conjecture on the value of $\bar{m}(\xi)$

where ξ is the limiting value of λ_1. Find an expression of ξ depending only on c, ϵ and $\|\mu\|_2$. Check your result with simulations.

Answer 8 We want to find ξ the asymptotic value of the dominant eigenvalue such that $\xi \in Sp(K_\epsilon)$ if there is a perturbation ($\mu \neq 0$) and $\xi \notin Sp(K_\epsilon)$ without perturbation ($\mu = 0$). I.e. $\bar{m}(\xi)$ is well defined in $\mathbb{C}$ and using eq. (1), $Det\left(I_n + \|\mu\|^2 \frac{\epsilon \bar{m}(\xi)}{c+\epsilon \bar{m}(\xi)} \frac{yy^T}{n}\right) = 0$. We then find:

$$\bar{m}(\xi) = -\frac{c}{\epsilon} \frac{1}{1 + \|\mu\|_2^2}$$

Now using eq. (2) we can recover the expression of ξ :

$$\boxed{\xi = 1 + \frac{\epsilon}{c}(1 + \|\mu\|^2) + \frac{1 + \frac{\epsilon}{\|\mu\|^2}}{1 + \|\mu\|^2}}$$

Regarding empirical testing for a hundred trials, with $n = 5000$ and $c = 0.8$, we computed that the computed ξ is kept in the 95% confidence interval of the empirical value. We deemed that metric sufficient to confirm our expression of ξ.

```python
def xi(c, epsilon, norm_mu):
    mu2 = norm_mu**2
    return 1 + (epsilon/c)*(1+mu2) + (1 + epsilon/mu2)/(1+mu2)


def check_xi(c=0.8, epsilon=0.5, norm_mu=3.0, n=2000, n_trials=30):
    lambda1_vals = [np.max(np.linalg.eigvalsh(model(n, c, epsilon, norm_mu)[0]))
                    for _ in range(n_trials)]

    mean = np.mean(lambda1_vals)
    ci = stats.t.interval(0.95, df=n_trials-1, loc=mean, scale=stats.sem(lambda1_vals))

    print(f"Empirical lambda_1: {mean:.6f}  95% CI: [{ci[0]:.6f}, {ci[1]:.6f}]")
    print(f"Computed xi0: {xi_mine(c, epsilon, norm_mu):.6f}  {'in CI' if ci[0] <=
    ↪  xi_mine(c, epsilon, norm_mu) <= ci[1] else 'outside CI'}")

check_xi(n=5000)
```

```
Empirical lambda_1: 7.356251  95% CI: [7.345233, 7.367269]
Computed xi0: 7.355556  in CI
```

Question 9 Describe an approach to prove the conjecture made in the previous question. This question will be evaluated on the relevance of the proposed strategy, not on your actual ability to prove the conjecture. In that regard, you are free to "guess" results and use numerical evidence.

Answer 9 We need to prove or explain why $\lambda_1 \in Sp(K_\epsilon)$ i.e. Q is not defined in λ_1 implies $\bar{Q}$ is also not defined in ξ the limit of λ_1. Indeed we can maybe proceed by an absurd, and suppose that $\bar{Q}(\xi)$ is well defined, then use isotrope convergence.

Our assumptions are the following: there is only one dominant eigenvalues outside the bulk, and this dominant eigenvalue λ_1 is converging toward ξ such that $\bar{Q}$ is not defined (diverging) in ξ.

Question 10 Relying on Cauchy's integral formula and the residue theorem, find an expression of the asymptotic value of $\frac{1}{n}|u_1^T y|^2$ depending only on c, ϵ and $\|\mu\|_2$. Check your result with simulations.

Answer 10 We consider a contour $\mathcal{C}^+$ exclusively enclosing the eigenvalue λ_1, the dominant eigenvalue of K_ϵ (possible because its separates from the bulk) and using the Cauchy's integral formula:

$$\frac{1}{2\pi i} \oint_{\mathcal{C}^+} Q(z)dz = [u_1 \cdots] \begin{bmatrix} \frac{1}{2\pi i} \oint_{\mathcal{C}^+} \frac{1}{\lambda_1 - z} dz & & \\ & \ddots & \\ & & \frac{1}{2\pi i} \oint_{\mathcal{C}^+} \frac{1}{\lambda_n - z} dz \end{bmatrix} \begin{bmatrix} u_1^\star \\ \vdots \end{bmatrix} = u_1 u_1^\star$$

Using eq. (1) and Sherman-Morrisson, we have:

$$\bar{Q}(z) = \bar{m}(z)\left[I - \left(1 - \frac{1}{1 + \|\mu\|^2 \frac{\epsilon\bar{m}(z)}{c+\epsilon\bar{m}(z)}}\right)\frac{yy^T}{n}\right]$$

With eq. (3) and a dominated convergence theorem , $\frac{1}{n}y^\star u_1 u_1^\star y \to \frac{1}{2\pi i}\oint_{\mathcal{C}^+} \frac{\bar{m}(z)}{1 + \|\mu\|^2 \frac{\epsilon\bar{m}(z)}{c+\epsilon\bar{m}(z)}} dz$.

And $\frac{\bar{m}(z)}{1+\|\mu\|^2 \frac{\epsilon\bar{m}(z)}{c+\epsilon\bar{m}(z)}} = \frac{(c+\epsilon\bar{m})\bar{m}}{\epsilon(1+\|\mu\|^2)} \frac{1}{\bar{m}(z)-\bar{m}(\xi)}$ thus the result is $\frac{\frac{c}{\epsilon}+\bar{m}(\xi)}{1+\|\mu\|^2}\frac{\bar{m}(\xi)}{\bar{m}'(\xi)}$.

And $\frac{(\frac{c}{\epsilon}+\bar{m}(\xi))\bar{m}(\xi)}{\bar{m}'(\xi)} = -[3(1-\epsilon)\frac{\epsilon}{c}\bar{m}^2 + 2\xi\bar{m} + \frac{c}{\epsilon}(\xi - 1) + 1] = -\|\mu\|^2 + \frac{\frac{c}{\epsilon} - \frac{c}{\|\mu\|^2}}{1+\|\mu\|^2} - \frac{\frac{c}{\epsilon} - 3c - \frac{2c}{\|\mu\|^2}}{(1+\|\mu\|^2)^2}$

$$\frac{1}{n}y^\star u_1 u_1^\star y \underset{n,d\to+\infty}{\longrightarrow} \boxed{-\frac{\|\mu\|^2}{1+\|\mu\|^2} + \frac{\frac{c}{\epsilon} - \frac{c}{\|\mu\|^2}}{(1+\|\mu\|^2)^2} - \frac{\frac{c}{\epsilon} - 3c - \frac{2c}{\|\mu\|^2}}{(1+\|\mu\|^2)^3}}$$

```python
def asymptotic_overlap(c, epsilon, norm_mu):
    mu2 = norm_mu**2
    t1 = mu2 / (1 + mu2)
    t2 = -(c/epsilon - c/mu2) / (1 + mu2)**2
    t3 = (c/epsilon - 3*c - 2*c/mu2) / (1 + mu2)**3
    return t1 + t2 + t3

def check_overlap(c=0.8, epsilon=0.5, norm_mu=3.0, n=2000, n_trials=30):
    vals = []
    for _ in range(n_trials):
        K_eps, y = model(n, c, epsilon, norm_mu)
        eigvals, eigvecs = np.linalg.eigh(K_eps)
        u1 = eigvecs[:, -1]
        vals.append((u1 @ y)**2 / n)

    mean = np.mean(vals)
    ci = stats.t.interval(0.95, df=n_trials-1, loc=mean, scale=stats.sem(vals))
    th = asymptotic_overlap(c, epsilon, norm_mu)

    print(f"Empirical 1/n * |u1^T y|^2: {mean:.6f}  95% CI: [{ci[0]:.6f}, {ci[1]:.6f}]")
    print(f"Theoretical value: {th:.6f}  {'in CI' if ci[0] <= th <= ci[1] else 'outside CI'}")

check_overlap()
```

```
Empirical 1/n * |u1^T y|^2: 0.883192  95% CI: [0.881965, 0.884419]
Theoretical value: 0.883911  in CI
```

Question 11 What is missing to find the performance of this spectral clustering algorithm? Propose an approach based on a realistic assumption

Answer 11 To find the performance, we need to have a value for y as we cannot quantify alignment without it. Let's assume that the clusters are balanced. Thus

$$\frac{1}{n}\|y\|^2 = 1, \qquad \frac{1}{n}\mathbf{1}^T y = \frac{n_+ - n_-}{n} = 0$$

so y is approximately orthogonal to $\mathbf{1}$.

Under our assumption, u_1 can be decomposed as

$$u_1 = \frac{y}{\sqrt{n}} \cdot \frac{u_1^T y}{\sqrt{n}} + u_1^\perp$$

where $u_1^\perp \perp y$. The component along y has magnitude $\frac{1}{\sqrt{n}}\sqrt{\frac{1}{n}|u_1^T y|^2}$, and the orthogonal part $u_1^\perp$ is distributed isotropically with variance $\frac{1}{n}(1 - \frac{1}{n}|u_1^T y|^2)$ per coordinate. Thus, asymptotically:

$$(u_1)_i \sim \mathcal{N}\left(\frac{y_i}{\sqrt{n}}\sqrt{\frac{1}{n}|u_1^T y|^2},\ \frac{1}{n}\left(1 - \frac{1}{n}|u_1^T y|^2\right)\right)$$

Using the algo proposed in **Question 6**, we define $\hat{y}_i = \text{sign}((u_1)_i)$, and the missclassification rate becomes

$$\boxed{\text{err} \underset{n,d\to+\infty}{\longrightarrow} \Phi\left(-\sqrt{\frac{\frac{1}{n}|u_1^T y|^2}{1-\frac{1}{n}|u_1^T y|^2}}\right)}$$

where $\Phi(x) = \int_{-\infty}^{x} d\mu(x)$ for $\mu \sim \mathcal{N}(0,1)$, and $\frac{1}{n}|u_1^T y|^2$ is given by the formula derived in **Question 10**.

```python
def check_u1_gaussian(c=0.8, epsilon=0.5, norm_mu=3.0, n=2000):
    K_eps, y = model(n, c, epsilon, norm_mu)
    eigvals, eigvecs = np.linalg.eigh(K_eps)
    u1 = eigvecs[:, -1]

    alpha = (u1 @ y)**2 / n
    theoretical_mean = np.sqrt(alpha) / np.sqrt(n)
    theoretical_std  = np.sqrt((1 - alpha) / n)

    u1_plus  = u1[y ==  1]
    u1_minus = u1[y == -1]

    print(f"Overlap 1/n * |u1^T y|^2: {alpha:.4f}")
    print(f"\nCluster +1:")
    print(f"  empirical mean: {np.mean(u1_plus):.6f} theoretical: {+theoretical_mean:.6f}")
    print(f"empirical std: {np.std(u1_plus):.6f} theoretical: {theoretical_std:.6f}")
    print(f"\nCluster -1:")
    print(f"  empirical mean: {np.mean(u1_minus):.6f} theoretical: {-theoretical_mean:.6f}")
    print(f"  empirical std: {np.std(u1_minus):.6f} theoretical: {theoretical_std:.6f}")

    u1_plus_standardized  = (u1_plus  - np.mean(u1_plus))  / np.std(u1_plus)
    u1_minus_standardized = (u1_minus - np.mean(u1_minus)) / np.std(u1_minus)

    ks_plus  = stats.kstest(u1_plus_standardized,  'norm')
    ks_minus = stats.kstest(u1_minus_standardized, 'norm')
    print(f"KS test cluster +1: stat={ks_plus.statistic:.4f}, p-value={ks_plus.pvalue:.4f}")
    print(f"KS test cluster -1: stat={ks_minus.statistic:.4f}, p-value={ks_minus.pvalue:.4f}")

def check_error_rate(c=0.8, epsilon=0.5, norm_mu=3.0, n=2000, n_trials=30):
    from scipy.special import ndtr

    empirical_errors = []
    for _ in range(n_trials):
        K_eps, y = model(n, c, epsilon, norm_mu)
        eigvals, eigvecs = np.linalg.eigh(K_eps)
        u1 = eigvecs[:, -1]
        y_hat = np.sign(u1)
        err = min(np.mean(y_hat != y), np.mean(y_hat != -y))
        empirical_errors.append(err)

    alpha = asymptotic_overlap(c, epsilon, norm_mu)
    theoretical_err = ndtr(-np.sqrt(alpha / (1 - alpha)))

    mean = np.mean(empirical_errors)
    ci = stats.t.interval(0.95, df=n_trials-1, loc=mean, scale=stats.sem(empirical_errors))

    print(f"Empirical error: {mean:.6f}  95% CI: [{ci[0]:.6f}, {ci[1]:.6f}]")
    print(f"Theoretical error: {theoretical_err:.6f}  {'in CI' if ci[0] <= theoretical_err <= ci[1] else 'outside CI'}")

check_u1_gaussian()
check_error_rate()
```

```
Overlap 1/n * |u1^T y|^2: 0.8826

Cluster +1:
  empirical mean: -0.020873 theoretical: 0.021007
empirical std: 0.007514 theoretical: 0.007662

Cluster -1:
  empirical mean: 0.021150 theoretical: -0.021007
  empirical std: 0.007813 theoretical: 0.007662
KS test cluster +1: stat=0.0171,  p-value=0.9196
KS test cluster -1: stat=0.0186, p-value=0.8853
Empirical error: 0.002700  95% CI: [0.002226, 0.003174]
Theoretical error: 0.002896  in CI
```

4 Discussion

Question 12 Discuss the sparsity assumption made here (keeping a random subset of the entries of K). Can you find a more clever policy for the choice of the entries of K to keep? What difficulties do you expect in the analysis of this model?

Answer 12 The sparsity assumption made here is not really realistic. Indeed, it is intuitive that keeping only a random subset of the Gram matrix's $K_\epsilon = K \odot B_\epsilon$ entries seems to result in a significant loss of information and therefore hinder the clustering process, unlike the use of a threshold to make the matrix sparse.

With this sparse model, $K_\epsilon = [K_{i,j} 1_{K_{i,j} \geq \epsilon}]_{i,j}$, and it should be possible to efficiently store and process K_ϵ, and even compute it's eigenvalues. However, using this threshold is highly non-linear and non-random, so it should be more difficult to analyse this model.